

## LANDER 11 SSE, WY, USA

WMO: 725730

Lat: 42.6754N

Lon: 108.6686W

Elev: 1760

StdP: 81.90

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94078

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -21.0      | -17.9 | -24.5                         | 0.5 | -20.2 | -21.5 | 0.7 | -16.6 | 12.3                  | 4.8  | 10.4 | 4.6  | 0.9                   | N/A  | 0.537 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.2                   | 33.0            | 13.4 | 31.5 | 13.2 | 30.1 | 13.1 | 15.6                | 27.1 | 15.0 | 26.6 | 14.3 | 26.0 | 3.4                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 12.0                            | 10.8 | 17.8 | 10.8 | 10.0 | 18.3 | 9.5 | 9.2 | 18.6 | 50.2          | 26.9 | 48.1 | 26.5 | 45.9 | 26.0 | 18.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.7           | 7.0  | 5.8 | -25.7                      | 35.4 | 2.4                | 1.1 | -27.4                                              | 36.2 | -28.8      | 36.9 | -30.1      | 37.5 | -31.8      | 38.3 |    |    |
| (5)               |      |     | -25.3                      | 17.4 | 2.2                | 0.7 | -26.9                                              | 17.9 | -28.2      | 18.3 | -29.4      | 18.6 | -31.0      | 19.1 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 7.9   | -4.3  | -3.0  | 2.8   | 6.2   | 10.9  | 17.5  | 22.4  | 20.9  | 15.9  | 7.8   | 1.7   | -4.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 10.64 | 5.95  | 6.32  | 5.44  | 4.91  | 4.96  | 4.31  | 3.04  | 3.25  | 4.76  | 5.57  | 5.92  | 6.10 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2034  | 444   | 363   | 227   | 132   | 48    | 3     | 0     | 0     | 9     | 103   | 251   | 454  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4098  | 702   | 597   | 481   | 364   | 233   | 64    | 6     | 16    | 99    | 326   | 498   | 712  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1278  | 0     | 0     | 5     | 18    | 77    | 229   | 385   | 338   | 187   | 36    | 3     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 299   | 0     | 0     | 0     | 0     | 3     | 41    | 132   | 96    | 27    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 3726  | 0     | 0     | 0     | 4     | 64    | 590   | 1607  | 1101  | 351   | 10    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1401  | 0     | 0     | 0     | 0     | 6     | 208   | 686   | 409   | 90    | 1     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.2   | 1.8   | 2.2   | 2.5   | 2.5   | 2.3   | 2.5   | 2.2   | 2.3   | 2.0   | 2.0   | 1.9   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 327   | 18    | 16    | 25    | 43    | 57    | 33    | 20    | 17    | 26    | 30    | 20    | 21   | (15) |
| (16) |                                                                                  | PrecMax     | 508   | 38    | 40    | 93    | 122   | 152   | 119   | 59    | 53    | 85    | 95    | 49    | 54   | (16) |
| (17) |                                                                                  | PrecMin     | 171   | 6     | 4     | 1     | 7     | 3     | 4     | 1     | 4     | 3     | 8     | 2     | 7    | (17) |
| (18) |                                                                                  | PrecStd     | 82    | 7     | 10    | 19    | 27    | 42    | 28    | 15    | 11    | 17    | 20    | 13    | 11   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 10.3  | 12.8  | 19.8  | 23.1  | 27.6  | 33.6  | 35.6  | 33.6  | 31.3  | 24.8  | 18.2  | 11.5 | (19) |
| (20) |                                                                                  |             | MCWB  | 3.0   | 4.3   | 7.2   | 9.0   | 10.7  | 13.1  | 13.8  | 13.4  | 13.3  | 10.6  | 7.1   | 3.9  | (20) |
| (21) |                                                                                  | 2%          | DB    | 8.0   | 10.2  | 16.6  | 20.1  | 25.4  | 31.3  | 33.8  | 32.0  | 29.1  | 21.5  | 15.3  | 8.8  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.1   | 3.3   | 6.1   | 7.9   | 10.9  | 12.8  | 13.6  | 13.0  | 11.9  | 9.1   | 5.6   | 2.5  | (22) |
| (23) |                                                                                  | 5%          | DB    | 6.2   | 8.1   | 14.0  | 17.6  | 22.7  | 29.2  | 32.1  | 30.7  | 27.4  | 19.0  | 13.1  | 6.9  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.1   | 2.1   | 4.9   | 6.9   | 10.2  | 12.5  | 13.6  | 12.8  | 11.3  | 8.2   | 4.8   | 1.8  | (24) |
| (25) |                                                                                  | 10%         | DB    | 4.4   | 6.1   | 11.6  | 15.4  | 20.1  | 26.9  | 30.6  | 29.2  | 25.2  | 16.9  | 10.8  | 4.6  | (25) |
| (26) |                                                                                  |             | MCWB  | 0.0   | 0.8   | 3.9   | 6.0   | 9.3   | 12.4  | 13.7  | 12.7  | 11.0  | 7.4   | 3.6   | 0.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 3.4   | 5.3   | 7.8   | 10.0  | 13.4  | 16.5  | 16.9  | 15.9  | 15.5  | 11.5  | 7.3   | 4.1  | (27) |
| (28) |                                                                                  |             | MCDWB | 8.3   | 11.3  | 18.5  | 22.1  | 22.2  | 26.7  | 27.6  | 26.2  | 27.3  | 21.7  | 16.8  | 9.3  | (28) |
| (29) |                                                                                  | 2%          | WB    | 2.4   | 3.9   | 6.5   | 8.5   | 11.8  | 15.1  | 15.8  | 15.0  | 13.6  | 10.2  | 5.8   | 2.9  | (29) |
| (30) |                                                                                  |             | MCDWB | 7.2   | 10.5  | 15.8  | 18.8  | 20.8  | 26.1  | 27.6  | 26.4  | 23.4  | 20.1  | 14.2  | 8.3  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.4   | 2.6   | 5.4   | 7.4   | 10.7  | 14.1  | 15.3  | 14.2  | 12.7  | 9.2   | 4.8   | 1.6  | (31) |
| (32) |                                                                                  |             | MCDWB | 6.1   | 8.4   | 13.8  | 16.6  | 19.8  | 25.4  | 27.1  | 26.1  | 23.5  | 18.1  | 12.2  | 6.6  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.1   | 1.2   | 4.3   | 6.4   | 9.8   | 13.2  | 14.7  | 13.7  | 11.8  | 8.0   | 3.6   | 0.1  | (33) |
| (34) |                                                                                  |             | MCDWB | 4.3   | 6.4   | 11.7  | 14.8  | 18.4  | 23.9  | 26.7  | 26.0  | 22.7  | 16.5  | 10.0  | 4.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 10.3  | 10.4  | 11.4  | 12.1  | 12.0  | 14.6  | 15.2  | 14.9  | 14.0  | 12.2  | 11.3  | 10.2 | (35) |
| (36) |                                                                                  |             | MCDBR | 12.2  | 12.5  | 14.3  | 15.6  | 15.6  | 17.1  | 16.7  | 16.6  | 15.9  | 15.1  | 13.9  | 12.8 | (36) |
| (37) |                                                                                  |             | MCWBR | 7.9   | 7.6   | 7.1   | 7.5   | 6.9   | 6.9   | 6.0   | 6.1   | 6.2   | 7.0   | 7.1   | 8.1  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 12.1  | 12.3  | 13.8  | 15.0  | 13.5  | 15.4  | 15.1  | 14.9  | 13.4  | 13.7  | 13.9  | 12.9 | (38) |
| (39) |                                                                                  |             | MCWBR | 8.1   | 7.6   | 7.1   | 7.3   | 6.7   | 7.2   | 5.9   | 5.7   | 5.4   | 6.5   | 7.5   | 8.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.234 | 0.240 | 0.260 | 0.305 | 0.317 | 0.313 | 0.340 | 0.331 | 0.298 | 0.265 | 0.244 | 0.235 | (40) |      |
| (41) |                                                                                  | taud        | 2.519 | 2.524 | 2.519 | 2.411 | 2.440 | 2.478 | 2.428 | 2.444 | 2.532 | 2.601 | 2.576 | 2.523 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 942   | 988   | 999   | 964   | 955   | 955   | 925   | 926   | 941   | 944   | 923   | 909   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 67    | 80    | 92    | 111   | 111   | 108   | 112   | 106   | 90    | 74    | 64    | 61    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.16  | 3.16  | 4.43  | 5.55  | 6.39  | 7.59  | 7.31  | 6.38  | 5.18  | 3.54  | 2.37  | 1.76  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.14  | 0.20  | 0.26  | 0.32  | 0.52  | 0.46  | 0.30  | 0.20  | 0.35  | 0.23  | 0.18  | 0.13  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (10 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page